

CS543/ECE549: Computer Vision

Quick links: [schedule](#), [lecture videos](#) (choose Log In Via Institution), [Piazza](#) (announcements and discussion), [Compass](#) (assignment submission and grades)



Instructor: [Svetlana Lazebnik](#) (slazebni-at-illinois.edu)

Lectures: T TH 11:00-12:15, 1310 DCL

Instructor office hours (3308 Siebel): Tuesdays 2-3PM, Thursdays 3-4PM

TAs: Lavisha Aggarwal (lavisha2), Hsiao-Ching Chang (hchang65), Wei Han (weihan3), Unnat Jain (uj2)

TA office hours (207 Siebel): Mondays 5-6PM, Wednesdays 12-1PM and 5-6PM, Fridays 1-2PM

Always check announcements on Piazza for short-notice changes to instructor and TA office hours!

Overview

In the simplest terms, computer vision is the discipline of "teaching machines how to see." This field dates back more than fifty years, but the recent explosive growth of digital imaging and machine learning technologies makes the problems of automated image interpretation more exciting and relevant than ever. There are two major themes in the computer vision literature: 3D geometry and recognition. The first theme is about using vision as a source of *metric 3D information*: given one or more images of a scene taken by a camera with known or unknown parameters, how can we go from 2D to 3D, and how much can we tell about the 3D structure of the environment pictured in those images? The second theme, by contrast, is all about vision as a source of *semantic information*: can we recognize the objects, people, or activities pictured in the images, and understand the structure and relationships of different scene components just as a human would? This course will provide a coherent perspective on the different aspects of computer vision, and give students the ability to understand state-of-the-art vision literature and implement components that are fundamental to many modern vision systems.

Prerequisites: Basic knowledge of probability, linear algebra, and calculus. MATLAB programming experience and previous exposure to image processing are highly desirable.

Recommended textbooks:

- [Computer Vision: A Modern Approach](#) by David Forsyth and Jean Ponce (2nd ed.)
- [Computer Vision: Algorithms and Applications](#) by Richard Szeliski (PDF available online)

Grading scheme:

- **Programming assignments:** 50%
 - Five MPs, done individually. Make sure you have access to MATLAB with the Image Processing Toolbox
- **Final project:** 30%
 - Groups of up to three; deliverables include proposal, intermediate progress report, final report. Selected groups may do a brief presentation for extra credit
- **Unit quizzes:** 20%
 - Four multiple-choice online quizzes on the four units from the syllabus below
- **Participation:** up to 3% extra credit
 - Students can get extra credit for actively participating in class (preferred), on Piazza, or during office hours

Academic integrity policy: You are encouraged to discuss assignments with each other, but coding and writing of reports must be done *individually* unless specifically instructed otherwise. You are also encouraged to search the Web for tips or code snippets, provided this does not make the assignment trivial and all external sources are explicitly acknowledged in the report. At the first instance of cheating (copying from other students or unacknowledged sources on the Web), a grade of zero will be given for the respective assignment or test. At the second instance, you will automatically receive an F for the entire course.

Syllabus

I. Image formation and low-level vision

- Camera models
- Light and color
- Linear filters and edges
- Feature extraction
- Optical flow and feature tracking

II. Grouping and fitting

- Hough transform
- RANSAC
- Alignment, image stitching

III. Geometric vision

- Camera calibration
- Epipolar geometry
- Two-view and multi-view stereo
- Structure from motion

IV. Recognition and beyond

- Statistical learning framework
- Image classification
- Deep convolutional neural networks
- Object detection
- Segmentation

Schedule (tentative)

Date	Topic	Readings (F&P 2nd ed.), assignments
January 16	Introduction: PPT , PDF	Homework: Assignment 0 Resource: MATLAB tutorial
January 18	Perspective projection: PPT , PDF	Reading: F&P ch. 1
January 23	Cameras: PPT , PDF	
January 25	Light and shading: PPT , PDF	Reading: F&P ch. 2 Assignment 0 due January 29, 11:59:59PM
January 30	Color: PPT , PDF	Reading: F&P ch. 3 Homework: Assignment 1 MATLAB , Python
February 1	Linear filtering: PPT , PDF	Reading: F&P ch. 4
February 6	Edge detection: PPT , PDF	Reading: F&P sec. 5.1-5.2
February 8	Corner detection: PPT , PDF	Reading: F&P sec. 5.3 Resource: Harris corner detector code
February 13	SIFT keypoints: PPT , PDF	Reading: Distinctive image features from scale-invariant keypoints

February 15	Optical flow: PPT , PDF	Reading: F&P sec. 11.1 Assignment 1 due February 19, 11:59:59PM
February 20	Fitting: PPT , PDF	Reading: F&P sec. 10.2-10.4, 22.1 Homework: Assignment 2 MATLAB , Python Project proposal
February 22	Hough transform: PPT , PDF	Reading: F&P sec. 10.1
February 27	Alignment: PPT , PDF	Reading: F&P sec. 12.1
March 1	Alignment cont.	
March 6	Camera calibration: PPT , PDF	Reading: F&P ch. 1
March 8	Single-view modeling: PPT , PDF	Reading: Ch. 2 from Hojem and Savarese book Assignment 2 and project proposals due March 12, 11:59:59PM
March 13	Epipolar geometry: PPT , PDF	Reading: F&P sec. 7.1 Homework: Assignment 3 is out
March 15	Binocular stereo: PPT , PDF	Reading: F&P ch. 7
March 27	Multi-view stereo: PPT , PDF	
March 29	Structure from motion: PPT , PDF	Reading: F&P ch. 8
April 3	Intro to recognition: PPT , PDF	
April 5	Recognition cont.	
April 10	Deep learning: PPT , PDF	
April 12	Deep learning cont.	Assignment 3 due April 12, 11:59:59PM Project progress report due April 16, 11:59:59PM
April 17	Detection: PPT , PDF	Homework: Assignment 4 is out
April 19	Detection cont.: PPT , PDF	
April 24	Segmentation: PPT , PDF	
April 26	CNNs for segmentation and beyond: PPT , PDF	
May 1	Selected project presentations	Assignment 4 due May 2, 11:59:59PM Final project reports due May 8, 11:59:59PM

Useful Resources

Tutorials, review materials

- [MATLAB tutorial](#) (via David Kriegman and Serge Belongie)
- **More MATLAB tutorials:** [basic operations](#), [programming](#), [working with images](#) (via Martial Hebert at CMU)
- [Linear algebra review](#) (via David Kriegman)
- [Random variables review](#) (via David Kriegman)

MATLAB reference

- [MATLAB guide from Mathworks](#)
- [MATLAB image processing toolbox](#)
- [Writing fast code](#)